



ITES

- Isokinetic control
- Constant sampling
- Velocity measurement
- Impactor sampling
- Sampling data saving
- No limit of memory all seconds data-storage
- Data import to Excel

The ITES is an automatic control unit for isokinetic sampling of particulate matter, gases, PM 2.5/10 and dioxin sampling.

The system consists of three separated units:

- an intelligent controller
- case with external sensor placed at the probe
- corrosive resistant frequency controlled pump.

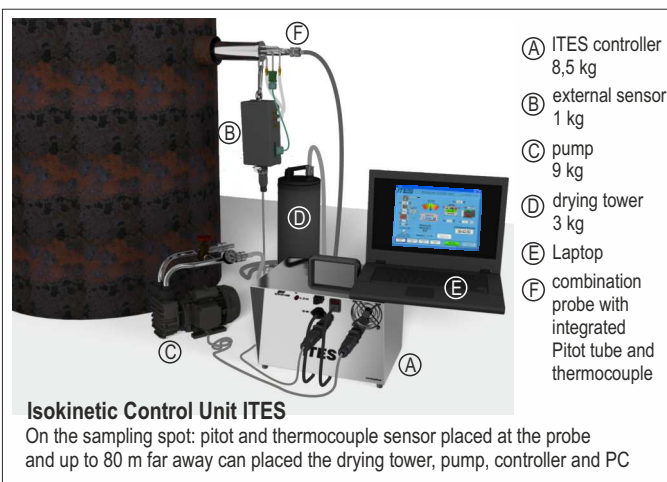
Operate with powerful gas tight pump.

Working range: 0.5 .. 4 m³/h

Suction rate at 800 hPa: 4 m³/h

Simple programming.

Sampling with filter devices according to EN, ISO and VDI Standards.



Isokinetic Control Unit ITES

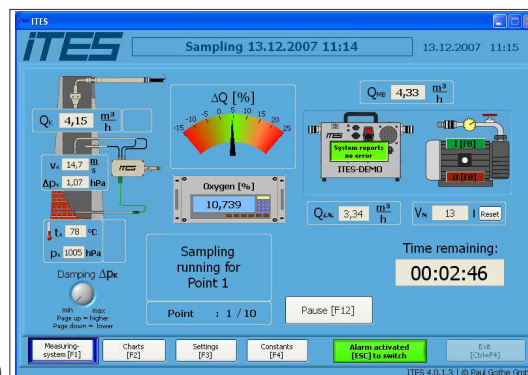
On the sampling spot: pitot and thermocouple sensor placed at the probe and up to 80 m far away can placed the drying tower, pump, controller and PC

ITES

- As supplement to existing systems
- With orifice technique
- Simple control menu
- With external alarm
- taking account of bypass flow
- Calculation sampling points
- Calculation density
- Calculation nozzle opening
- Calculation gas humidity

All necessary data stored in data file and can directly open in spreadsheets:

- Difference pressure of the Pitot tube
- Gas velocity
- Volume flow rate nozzle and orifice
- Calculated isokinetic deviation
- Temperature of the duct
- Pressure of the duct
- Sampling point
- Time of recording
- Sampling volume (total and each point)



Complete software controlled from standard Laptop
Developed for Windows 7, 8 and 10

ITES now with humidity measurement:

With one Psychrometer can measure directly the gas humidity.
Measurement processed automatically! With data report.

- Gas humidity [g/m³ and %]
- Dew point [°C/°F]

Technical Features

difference pressure sensor: 0,01..10 hPa	size: 350 x 280 x 250 mm
absolute pressure sensor: 40..130 kPa	alarm output: 12 V
temperature duct NiCr-Ni: 0 ..500°C	supply: 230 VAC, 50 Hz
operating temperature: -10..60°C	weight ITES: 8,5 kg
Siemens frequency converter	external sensor 1 kg
orifice range: 0.5 .. 4 m ³ /h	pump (5.3 or 8.8 m ³ /h)
stable steel casing	Art.-No: 25.10

Advantage of the ITES

- not „all in one“
ideal for fast replacement
- short cable from
probe to sensor
- report direct to PC
- low weight because
heavy pump separate
- sensor with different
ranges available



Control unit small and light, makes it easy portable, important if the access is very small. Special to whom wish want to carry only the control panel on stack platform and leave the pump on plant floor. Because of the high quality and last generation electronic-modules inside the control unit it is simplified the maintenance.

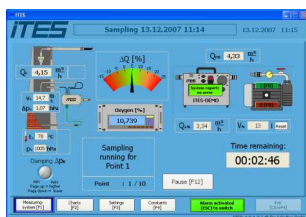
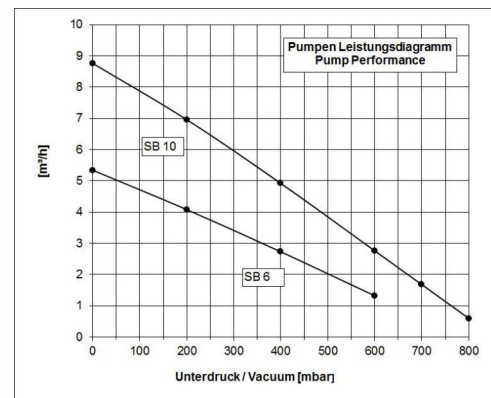


Separate sensor box for difference-, absolute pressure and temperature. The separate box can fixed at the end of the suction tube and allow a very short pressure line and the control unit with pump can place far away. Different sensor boxes can connect with the ITES to find individual solutions for special applications. Sensor transmitter can be programmed to the desired range.



Gas-tight dry running rotary vane vacuum pump. Particularly little maintenance requirements, resistant and highly efficient, 230V, more than 700 mbar vacuum, the pump is gas-tight, with vacuum manometer (-1 to 0 bar). Special edition with corrosion resistant pump body. Protection rating IP 55.

Two pumps available:
SB 6, range: 0 .. 5 m³/h, weight: 10 kg
SB 10, range: 0 .. 8 m³/h, weight: 15 kg

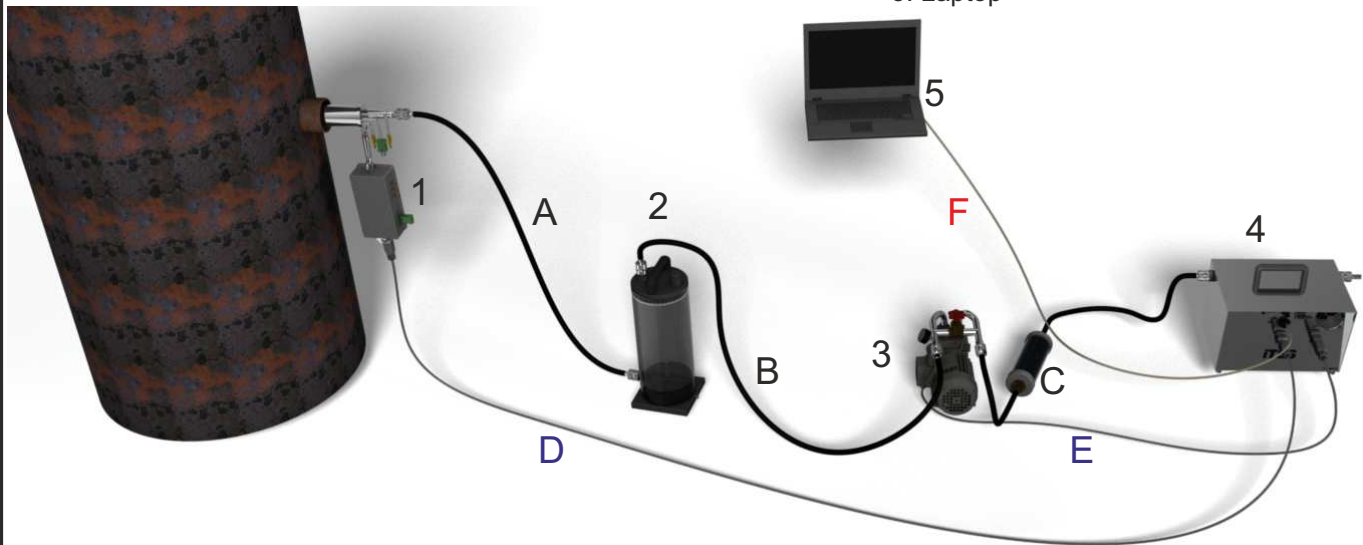


Parameters and data saved directly on PC. In case of power shut down, the data are save! Stored report type: Detailed measurement report with punctual report and sampled point summary and second short report with only the average data for fast check of the data. Report format compatible with Microsoft Excel, SUN Open Office suite or any other spread sheets. Unlimited free update of the software. Every update contains more functionality. The data processing thanks to PC controlling is unlimited.

- Main Characteristics:**
- all parameter can set manual important if it all goes wrong.
 - the software allow to manage the sampling in an easy and intuitive manner.
 - in accordance with UNI EN, US EPA and VDI
 - fast isokinetic control at any stack condition.
 - suction volume rate measurement and velocity measurement.
 - In-stack temperature and anomalies management.
 - autotest and anomalies management.
 - auto check at starting
 - leak test according EN
 - easy to accreditate

Verbindungen ITES: Mögliche Varianten Connections ITES: Possible variants

- 1: Sensor box: 200 x 60 x 100 mm, 1 kg
2: Drying Tower: Ø 160 x 400 mm, 3 kg
3: Pump: 250 x 180 x 320 mm, 10 kg
4: ITES: 350 x 280 x 250 mm, 8,5 kg
5: Laptop



- A: Absaugschlauch: Sonde - Trockenturm (Länge frei wählbar)
B: Absaugschlauch: Trockenturm - Pumpe (Länge frei wählbar)
C: Absaugschlauch: Pumpe - Pulsationsdämpfer - ITES (Länge frei wählbar)

- D: Kabel: Signalausgang Sensoren (Staurohr - 4..20 mA), Länge bis zu 100 m
E: Kabel: Frequenzsteuerung Pumpe: Länge max. 3 m

- F: RS 485 Kabel: ITES- Laptop: Länge bis zu 100 m

- A: Suction hose: Probe - Drying Tower (length can select free)
B: Suction hose: Drying Tower - Pump (length can select free)
C: Suction hose: Pump - pulsation damper - ITES (length can select free)

- D: Cable: Signal Output Sensor (Pitot Tube - 4..20 mA), Length up to 100 m
E: Cable: Frequency Controlled Pump: Length max. 3 m

- F: RS 485 Cable: ITES- Laptop: Length up to 100 m

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